SmartBoard Session Report

Course: Cpe413 - Micro
Session: Intro to Microchip
Date: 2026-10-23
Status: finalized

a. UTP b. Broadband c. Fibr d. Bluetooth

29. A set of high-speed networks that carry Internet traffic. These networks are provided by companies such as AT&T, GTE, and IBM.
a. Internet backdoor b. Internet backbone c. Internet Provider d. Trunk lines

30. A computer that stores and manages files for multiple users on a network.
a. Web Server b. File Server c. Virtual Server d. All of these

31. A collection of computing devices that are connected in various ways in order to communicate and share resources
a. Computer Network b. Internet c. IRC d. SMTP

32. A switching technique used by telecommunication networks that provides functionality that is similar to both circuit switching and packet switching networks.
a. Circuit Switching b. Packet Switching c. Frame Relay d. ATM

33. A client computer that is used to run applications and is connected to a server from which it obtains data shared with other computers.
a. Web Server b. File Server c. Workstation d. Personal Computer

34. It is an identification string that defines a realm of administrative autonomy, authority or control within the Internet.
a. Domain b. Packets c. IP d. HTTP

35. Which of the following is not a consideration of NOS.?
a. Robustness b. Security c. Performance d. Balance

36. Which of the following is not a TCP/IP based service for Internet mail?
a. POP3 b. IMAP c. SMTP d. HTTP

37. A logical subdivision of an IP network and the practice of dividing a network into two or more.
a. Subnet b. Access point c. Partitioning d. Networking

38. Subneting is a process of dividing large network into the smaller networks based on layer _ IP address?
a. Layer 1 b. Layer 2 c. Layer 3 d. Layer 4

39. PLDT, Digitel, Globe, Smart are examples of
a. Business b. Customer c. ISP d. HTTP

40. Who owns the Internet?
a. Bill Gates b. Steve Jobs c. Mark Zuckerberg c. None

41. A printed circuit board that fits into the expansion slot on the motherboard or peripheral device of a computer which is sometimes called as network adapter.
a. PCI b. AGP c. NIC d. UTP

42. It occurs when two bits propagate at the same time on the same network
a. Collision b. Broadcast c. Noise d. None of these

43. It is the amount of data that is transmitted at a particular period of time.
a. Throughput b. Bandwidth c. OSI d. Signal

44. It is defined as undesirable signals and can originate from natural and technological sources and is added to the data signals.
a. Wave b. Noise c. Jitter d. Codec

45. A template or reference model consisting of seven layers that define the operations performed on a computer network created by the ISO.
a. Computer Network b. Data Communication c. OSI Model d. Internet model

46. For a network using the IP address 172.16.0.0 /20, what is the broadcast address for the first subnet?
a. 172.16.0.255 b. 172.16.15.255 c. 172.16.31.255 d. 172.16.255.255

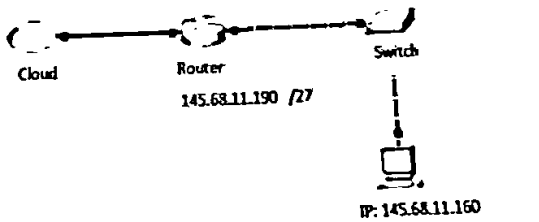
47. A device that modulates digital data onto an analog signal for transmission over a telephone line, then modulates the analog signal back to digital data.
a. Router b. Modem c. Hub d. Codec

48. A network hardware that is sometimes called unintelligent device because of the collision that always exist.
a. Hub b. Router c. Bridge d. Repeater

49. Which of the following is a class B subnet?
a. 192.168.0.0 b. 173.105.33.0 c. 192.0.0.0 d. 192.168.0.1

50. A network is assigned the subnet mask 255.255.255.248. How many usable hosts are available per subnet?
a. 6 b. 14 c. 30 d. 62

II – Subnetting : The computer cannot connect in the internet. Find the error and write recommendations of the subnet.



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Final Exam in Digital Signal Processing

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

b) Express X(z) in polynomial form.

c) Determine the zeros of X(z).

d) Determine the poles of X(z), if any.

e) Determine the region of convergence (ROC) of X(z).

3. Given:

$$X(z) = \frac{1}{(1-0.5z^{-1})(1-0.25z^{-1})}$$

with the region of convergence:

$$|z| > 0.5$$

Answer the following:

a) Express X(z) in partial fraction form.

b) Determine the inverse Z-transform x[n].

c) Calculate the first five samples: x[0], x[1], x[2], x[3], and x[4].

d) Determine whether the resulting sequence is right-sided, left-sided, or two-sided.

e) Determine whether the sequence is absolutely summable. Justify your answer.

4. Consider:

$$X(z) = \frac{z}{z-3}$$

Determine the inverse Z-transform for each of the following regions of convergence:

- Case 1: $|z| > 3$
- Case 2: $|z| < 3$

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Final Exam in Digital Signal Processing

II. Problem Solving. Show your complete solution and box your final answer for each problem.

1. A continuous-time signal contains frequency components up to 6 kHz. The signal is sampled at 10 kHz and quantized using an 8-bit analog-to-digital converter (ADC) with an input range of −4 V to 4 V.

Perform the following:

a) Determine the Nyquist sampling rate of the signal.

b) Determine whether the given sampling frequency satisfies the Nyquist–Shannon sampling theorem. Explain your answer.

c) Calculate the sampling period using:

$$T_s = \frac{1}{f_s}$$

d) Determine the total number of quantization levels using:

$$L = 2^b$$

e) Calculate the quantization step size using:

$$\Delta = \frac{V_{max} - V_{min}}{L}$$

f) Assuming a uniform mid-rise quantizer, determine the reconstruction levels and corresponding 8-bit binary codes for the first four quantization intervals, starting from the minimum input voltage.

2. Given the discrete-time signal:

$$x[n] = \{2,1,3,4\}, \quad 0 \le n \le 3$$

and x[n] = 0 otherwise, answer the following:

a) Determine the Z-transform X(z) using the definition:

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